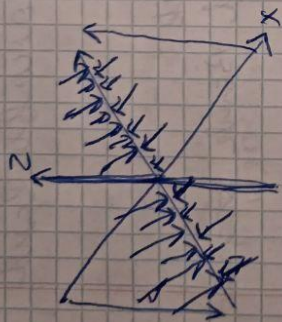


page 1085 ex 108

$$F(x, y, z) = x^2$$



x, y, z	$F(x, y, z)$
0, 0, 0	0, 0, 0
0, 0, 1	0, 0, 0
0, 1, 0	0, 0, 0
1, 0, 0	1, 0, 0
2, 0, 0	2, 0, 0
-1, 0, 0	0, 0, -1

end

$F(x, y, z)$ depends only on x

I took points where $x=1, -1, 2, 5, -2, 5$

Page 1086 ex 11

$$\int_C y^3 ds \quad C = \begin{cases} x=t^3 \\ y=t \\ z=t^3 \end{cases} \quad t \in [0, 2]$$

$$ds = \sqrt{(t^3)^2 + (1)^2 + (t^3)^2} dt = \sqrt{2t^6 + 1} dt$$

$$\int_0^2 t^3 \sqrt{2t^6 + 1} dt = \int_0^2 t^3 \sqrt{2t^6 + 1} dt$$

$$m_1 = 1$$

$$m_2 = 145$$

$$t^3 dt = \frac{dt^4}{4}$$

$$\int_1^{\sqrt{2}} \frac{dw}{\sqrt{w}} = \frac{1}{\sqrt{2}} \cdot \frac{2}{3} = \frac{1}{3\sqrt{2}} (\sqrt{2} - 1)$$

[page 1086, ex 15]

$$\int_0^1 (x^2 y^3 - \sqrt{y}) dy \quad y = \sqrt{x} \quad x \in [1, 4]$$

$$x = t$$

$$y = \sqrt{x} \quad t \in [1, 2]$$

$$dy = \frac{1}{2} \cdot \frac{1}{\sqrt{x}} dt = \frac{dt}{2\sqrt{t}} = \frac{1}{2} t^{-1/2} dt$$

$$\int_1^2 (t^2 \cdot t^{3/2} - t^{1/2}) \frac{1}{2} t^{-1/2} dt = \frac{1}{2} \int_1^2 (t^{5/2} - 1) dt$$

$$= \frac{1}{2} \left[\frac{2}{7} t^{7/2} - t \right]_1^2 = \frac{1}{2} \left(\frac{2}{7} \cdot 2^{7/2} - 2 - \left(\frac{2}{7} - 1 \right) \right)$$

$$= \frac{1}{2} \left(\frac{2}{7} \cdot 8\sqrt{2} - 2 - \left(-\frac{5}{7} \right) \right) = \frac{1}{2} \left(\frac{16\sqrt{2}}{7} - \frac{9}{7} \right)$$

[page 1086, ex 14]

$$\int_0^1 \int_0^1 \int_0^1 (x + y + z) dx dy dz$$

$$F(x, y, z) = \frac{1}{2} x^2 + x y + x z$$

$$F(1, 1, 1) = \frac{1}{2} + 1 + 1 = \frac{5}{2}$$

$$F(0, 0, 0) = 0$$

$$\begin{aligned}
 \int_C \vec{F}(x, y, z) \cdot d\vec{r} &= \int_0^1 (t^2 + t + 2t) dt \\
 &= \int_0^1 (t^2 + t + 2t) dt = \int_0^1 (t^2 + 3t) dt \\
 &= \left[\frac{t^3}{3} + \frac{3t^2}{2} \right]_0^1 = \frac{1}{3} + \frac{3}{2} = \frac{11}{6}
 \end{aligned}$$

page 1106 ex 15

a)

$$\begin{aligned}
 F(x, y, z) &= yz \cdot \vec{i} + xz \cdot \vec{j} + (xy + 2z) \cdot \vec{k} \\
 \text{integrated:} \\
 f'_x &= yz & f &= xyz + C \\
 f'_y &= xz & f &= xyz + C \\
 f'_z &= xy + 2z & f &= xyz + z^2 \\
 C^2 &= z^2 & f(x, y, z) &= xyz + z^2
 \end{aligned}$$

8) FTLLI

$\nabla f \Rightarrow \text{integral}$

$$\int_C \nabla f \cdot d\vec{r} = f(x_2, y_2, z_2) - f(x_1, y_1, z_1) = (4, 6, 3 + 9) - (1, 0, -2) = 77$$

$$f(x, y, z)$$

$$+ z^2 +$$

$$\int_2^4 \left(\frac{y}{3} \right) dy =$$

$$[xy + 2z]$$

$$+ 2z$$